

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2022–23

Paper: GEN-401 — Real Analysis

Time: Three Hours

Maximum Marks: 70

Instructions: This question paper carries three sections: Section A, B, and C. All questions are compulsory.

Section A — Long Answer Questions

[12 × 2 = 24]

Attempt any two questions. Each question carries 12 marks.

Question 1. Calculate the divergence and curl of the vector field $\mathbf{F} = x^2 \hat{i} + xy \hat{j} + yz \hat{k}$. Interpret the physical meaning of these operators.

Question 2. Prove that $f(x) = 2x + 3$ is continuous at $x = 1$.

Question 3. Find the first and second order partial derivatives of the function $f(x, y) = 3x^2 + 4xy - 5y^2$.

Question 4. Calculate the gradient of $f(x, y, z) = x^2 + xy + z^2$ and evaluate it at the point $(1, 1, 1)$.

Section B — Short Answer Questions

[6 × 6 = 36]

Answer any six out of the following nine questions. Each question carries 6 marks.

Question 5. Compute the curl of the vector field $\mathbf{G} = \sin(x) \hat{i} + \cos(y) \hat{j}$.

Question 6. Prove that $f(x) = x^2$ is continuous but not uniformly continuous on $\mathbb{R}$.

Question 7. Calculate the directional derivative of $f(x, y) = x^2 + y^2$ at the point $(1, 1)$ in the direction of $\hat{i} + \hat{j}$.

Question 8. Prove that the function $f(x, y) = x^2$ is differentiable at $x = 0$.

Question 9. Use the limit definition of a derivative to find $f'(x)$ for $f(x) = x^2$.

Question 10. Compute $\frac{\partial^2}{\partial x \partial y}(3x^2 + 4xy - 5y^2)$.

Question 11. Calculate the unit vector perpendicular to both $\mathbf{A} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\mathbf{B} = \hat{i} + \hat{j} - \hat{k}$.

Question 12. Compute the limit $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$.

Question 13. Show that the function $f(x) = x^2$ is not uniformly continuous on $\mathbb{R}$.

Section C — Very Short Answer Questions

[1 × 10 = 10]

Answer all ten questions. Each question carries 1 mark.

Question 14. What is the divergence of $\mathbf{F} = x\hat{i} + y\hat{j} + z\hat{k}$?

Question 15. True or False: Every continuous function is differentiable.

Question 16. What is the range of the function $f(x) = x^2$?

Question 17. True or False: $f(x) = \sin(x)$ is uniformly continuous on $\mathbb{R}$.

Question 18. What is the value of $\lim_{x \rightarrow 0} \frac{\sin x}{x}$?

Question 19. What is the cross product of $\mathbf{A} = \hat{i} + \hat{j}$ and $\mathbf{B} = \hat{j} - \hat{k}$?

Question 20. True or False: If f and g are continuous functions, then f/g is also continuous wherever $g \neq 0$.

Question 21. What is the curl of $\mathbf{F} = x^2\hat{i} + xy\hat{j} - z\hat{k}$?

Question 22. True or False: If a function is differentiable, it is continuous.

Question 23. True or False: The dot product of two vectors is a scalar.